

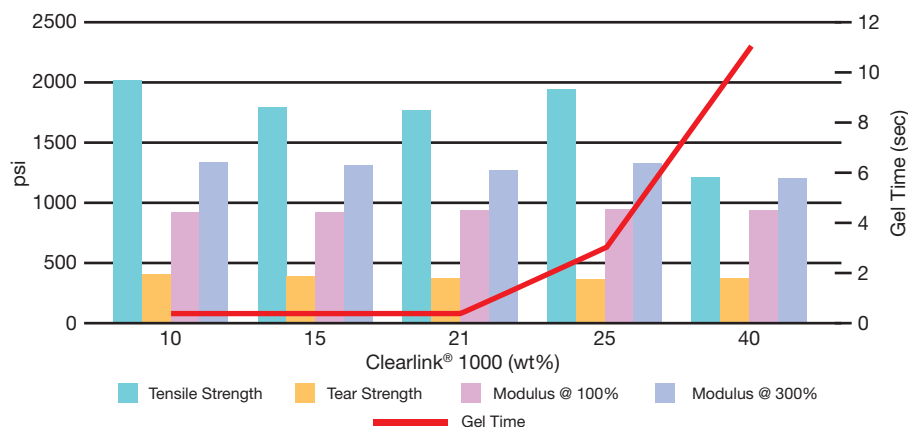
APPLICATIONS

POLYUREA COATINGS SYSTEMS

Clearlink® 1000 is a liquid aliphatic chain extender for light-stable or color-critical aliphatic polyurea spray and hand-applied coatings. It is a colorless liquid with low reactivity that provides considerable latitude for cure-rate control without altering formulation color, and it is compatible with a wide range of polyols, co-curing agents and other polyurethane and polyurea chemicals.

Clearlink® 1000 extends pot life, allowing time for coatings to flow, adhere and develop smooth finishes. Coatings cured with Clearlink® 1000 are tough and extremely light-stable in harsh environments.

Tensile Strength, Tear Strength, Modulus and Gel Time
Sprayed Polyurea Coating with Clearlink® 1000



Effect of Clearlink® 1000 on Properties of a Sprayed Polyurea Coating

A Side, wt%

IPDI: Jeffamine D-2000 (45:55)	100	100	100	100	100
NCO, %: 14.8					

B Side, wt%

Jeffamine D-2000	50.1	48.2	45.9	44.3	38.5
Jeffamine T-5000	19.2	18.5	17.6	17.0	14.8
IPDA	20.6	18.3	15.9	13.5	6.7
CLEARLINK® 1000	10.1	15.0	20.6	25.2	40.0
Gel Time (sec)	< 1	< 1	< 1	3	11
Tack Free (sec)	< 1	< 1	< 1	3	20

Physical Properties

Tensile Strength (psi)	2005	1810	1790	1945	1230
Elongation (%)	530	480	505	545	490
Tear Strength (psi)	420	395	380	375	380
Shore D Hardness	42	43	46	46	48
Modulus @ 100% (psi)	935	935	950	960	950
Modulus @ 300% (psi)	1350	1330	1305	1340	1220

HOW TO USE

Typical use levels in sprayed polyurea coatings range from 5% to 10% by weight.

DESCRIPTION

Clearlink® 1000 is an aliphatic amine chain extender for use with aliphatic isocyanates to produce color-critical and UV light-stable polymers. It is an easy-handling, low-viscosity liquid that is low in toxicity (Ames negative) and it provides substantially lower reactivity than other commercially available aliphatic diamines. Finished coatings have excellent color stability, physical properties and heat resistance.

TYPICAL PROPERTIES

PROPERTY	TYPICAL VALUE
Appearance	Clear, colorless liquid
Specific gravity @ 20 °C (68 °F)	0.90
Density, g/cm ³ (lb/gal)	0.89 (7.4)
Flash point, PMCC, °C (°F), min	60.6 (141)
Viscosity @ 15.6 °C (60 °F), cps	110
Water content, ppm	< 500
Molecular weight (g/mol)	322
Equivalent weight, min	161
Hydroxyl number	348
Ames test	Negative

PACKAGING AND SHIPPING

Not regulated for transportation by US DOT

SAFETY AND HANDLING

Please refer to the current Material Safety Data Sheet for detailed safety, handling and toxicity information.

TYPICAL SHELF LIFE

1 year



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